



COMPUTATIONAL LIFE SCIENCES / RESEARCH NOTE

AlphaGenome Atlas in RNU4-2: a focused computational benchmark

Variant ranking against published experimental cell-fitness measurements

We compared AlphaGenome Atlas AVI scores with published saturation genome editing measurements for 435 single-nucleotide variants across the 145-base RNU4-2 transcript. AVI showed a weak positive rank correlation with experimental cell-fitness loss (Spearman $r = 0.144$). CADD 1.7 scored 0.088. The paired difference interval included zero. Within-position allele concordance was 49.2%, close to its 50% chance expectation. These results provide no clear evidence of better ranking than CADD in this assay. This focused result is consistent with a limitation already discussed by the Atlas authors [2, p.18].

DATE / VERSION / AUTHOR

9 September 2026 UTC / 1.3 / NaS Research

RESEARCH NOTE

One RNA and one experimental endpoint; no clinical validation. Editorial revision 1.3, September 27, 2026. Numerical results and methods are unchanged.

Results

Metric	AVI	CADD 1.7	AVI minus CADD
Spearman correlation	0.144 [0.007, 0.260]	0.088 [-0.045, 0.225]	0.056 [-0.033, 0.137]
Within-position concordance	49.2% [44.1, 54.3]	45.5% [40.1, 50.7]	3.7 [-3.7, 10.9] pp

Brackets: 95% position-cluster bootstrap intervals. pp: percentage points. Primary analysis: 435 transcript SNVs at 145 positions.

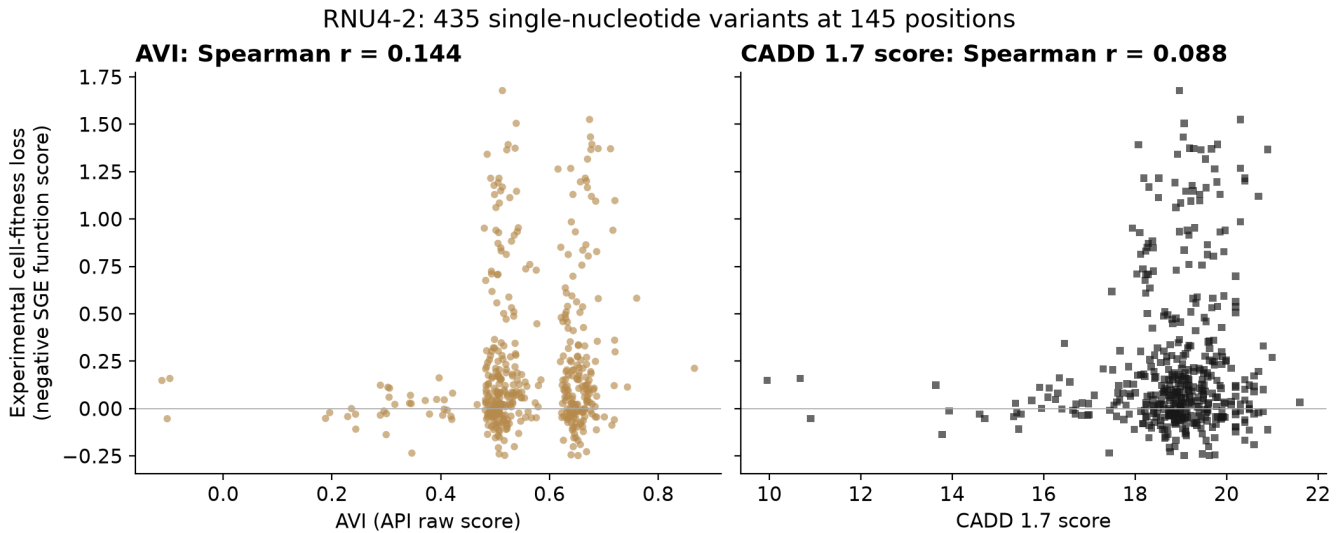


Figure 1. Scores versus experimental cell-fitness loss. Each point is one transcript SNV. AVI uses the raw API score; correlation is not accuracy.

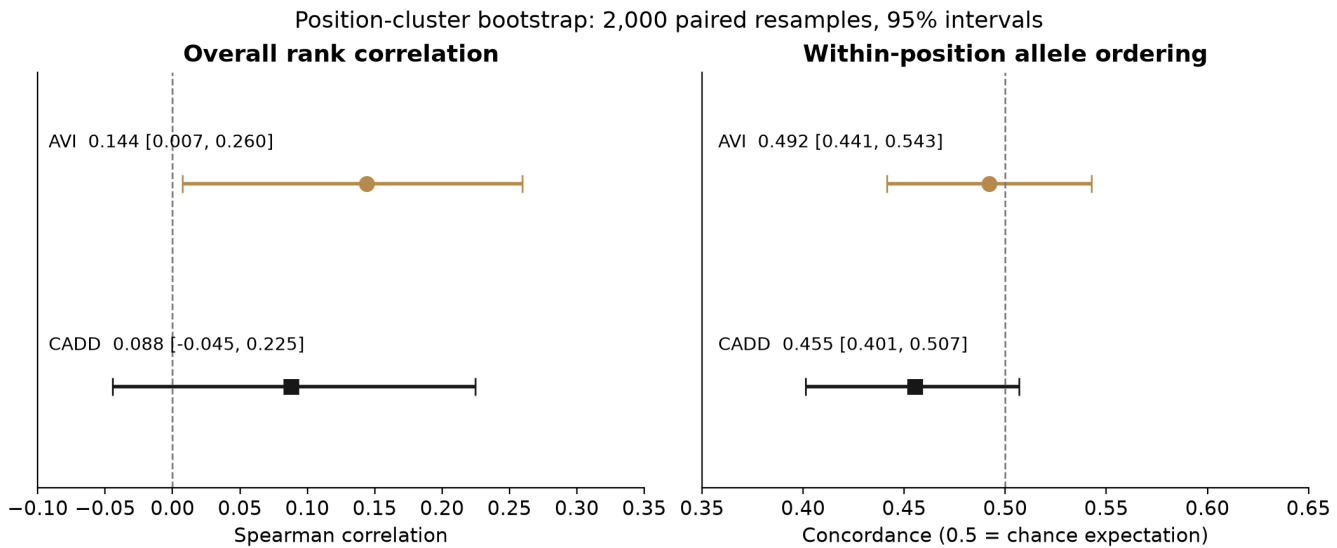


Figure 2. Estimates and 95% bootstrap intervals. The within-position statistic compares pairs of alternate alleles; 0.5 is its chance expectation.

Question and methods

Does AVI track measured variant effects in a non-protein-coding RNA, and does it distinguish alternate changes at the same position? Overall correlation can reflect differences between genomic regions. Within-position ordering asks a narrower question about allele discrimination.

Data and methods

We used the final published Supplementary Table 1 from De Jonghe et al. (2026) [1]. Of 539 engineered edits, 453 were SNVs. The primary cohort comprised all 435 transcript SNVs; 18 downstream SNVs were reserved for a prespecified sensitivity analysis and 86 other edits were excluded. Every forward-strand hg38 reference allele matched the UCSC reference sequence [4]. All 453 AVI scores were retrieved through the authenticated official Atlas client on 9 September 2026 UTC; no scores were missing. The endpoint was negative `function_score_mean`, representing cell-fitness loss. The comparator was CADD 1.7 from the same source table. We used the API raw AVI score, not a clinical or PHRED cutoff. Spearman correlation is invariant to strictly increasing score transformations.

Uncertainty and allele ordering

For the within-position metric, we compared all three pairs of alternative alleles at each site. Correct ordering scored 1, reversed ordering 0, and prediction ties 0.5; pairs tied experimentally were excluded. Site averages received equal weight. We used 2,000 paired bootstrap resamples of positions, preserving each site's alternate alleles, with seed 20260909. Intervals are percentile 95% intervals conditional on this one assay. We did not tune scores, fit a model, select a clinical threshold, or optimize the cohort against results.

Sensitivity analysis

Including all 453 SNVs yielded AVI $r = 0.172$ and CADD $r = 0.120$. The paired difference was 0.052 [-0.028, 0.131]. This did not materially change the interpretation. Secondary all-SNV allele-ordering estimates are retained in results.json but are not promoted as a separate confirmatory finding.

Interpretation and limits

The correlation of 0.144 describes rank agreement; it is not 14.4% accuracy. Similarly, 49.2% is a within-site pair-ordering statistic, not clinical classification accuracy. The data suggest limited rank agreement for this RNA and experimental endpoint. The near-chance allele ordering highlights a limitation that an overall correlation alone could obscure. However, tiny differences in experimental measurements can reverse allele order; our ordering metric does not incorporate replicate-level measurement uncertainty. Nearby sites may also be dependent, beyond the clustering captured by the bootstrap. HAP1 cell fitness is not patient severity, clinical pathogenicity, or performance across all non-coding variants. Failure to establish an advantage is not proof that the methods are equivalent.

Context and reproducibility

Relationship to previous work

Google's Atlas preprint explicitly identifies RNU4-2 as an example of a training-data gap: standard poly(A)-selected RNA sequencing excludes non-polyadenylated noncoding RNAs [2, p.18]. This is a plausible context for our result, not a mechanism established by this analysis. The authors report the highest Spearman correlation in eight of ten held-out SGE screens [2, p.5]; strong aggregate performance does not imply equal performance in every gene or assay. They already evaluate RNU4-2 [2, Methods p.28]. Our work is a focused reanalysis of an existing screen, not a first benchmark or new independent experimental validation. We used the final 2026 source table and did not reproduce Google's exact filtered input snapshot. The analysis plan was internally frozen before AVI retrieval, not externally preregistered. We do not establish that Atlas is broadly inaccurate or contradict Google's aggregate performance claims.

Reproducibility and review

Deterministic code produced all numerical results. Independent formulations reproduced the point estimates; source-value comparisons, input hashes, and exact bootstrap replay passed. Public experimental data and the free non-commercial API were used. No patient-level records or paid data services were required. A separate computational audit checked every source field, score, coordinate, estimate, and interval. No clinical recommendation is made. Reproduction files accompany the web edition.

References and attribution

- [1] De Jonghe et al. (2026). Saturation editing of RNU4-2 reveals distinct dominant and recessive disorders. Nature.
- [2] Cheng et al. (2026). AlphaGenome Atlas preprint, Discussion p.18; evaluation pp.5 and 28.
- [3] Google DeepMind. AlphaGenome Atlas API reference.
- [4] UCSC Genome Browser API. hg38 reference, chr12:120291752-120291903 (0-based half-open).
- [5] AlphaGenome Output Terms of Use.

Experimental source: De Jonghe et al., CC BY 4.0, subject to source exceptions. Figures are NaS derivatives. AlphaGenome outputs are subject to applicable Google terms. No credentials are included.